

Glossary: Networking Fundamentals

Welcome! This alphabetized glossary contains many of the terms you'll find within this course. This comprehensive glossary also includes additional industry-recognized terms not used in course videos. These terms are important for you to recognize when working in the industry, participating in user groups, and participating in other certificate programs.

Term	Definition
Address Resolution Protocol (ARP)	A protocol that deals with troubleshooting network issues and setting up new devices.
Bridge	A networking device that connects two or more separate computer networks, allowing them to function as a single network.
Clients	Devices that access the server are known as clients.
Domain Name System (DNS)	A network protocol that resolves or translates domain names to IP addresses.
Dynamic Host Configuration Protocol (DHCP)	A network protocol that automates IP address and configuration assignment for devices on a network.
File Transfer Protocol (FTP)	A protocol responsible for file transfer.
Firewall	A hardware or software that monitors and controls incoming and outgoing network traffic based on predetermined security rules.
Gateway	A hardware or software that allows data to flow from one network to another. Gateways use multiple protocols to connect multiple networks.
Hub	A networking device that connects multiple devices together in a network. When data is sent to a hub, it broadcasts the data to all network devices except the sender.
Hypertext Transfer Protocol (HTTP)	A protocol responsible for standard Internet or web page access.
Hypertext Transfer Protocol Secure (or HTTPS)	A protocol responsible for encrypted Internet or web page access. HTTPS should be used on a web page that prompts for sensitive data such as credit card numbers.
Internet Message Access Protocol version 4 (IMAP4)	A protocol responsible for receiving emails. Email is stored on the server and synchronized to multiple devices, like laptops and mobile phones.
Internet of Things (IoT) network model	A network model that comprises diverse communication protocols, including data collection, data package, data transfer, and data control.
Intrusion Detection System (IDS)	A security tool that monitors network or system activity for malicious activity or policy violations.
Intrusion Prevention System (IPS)	A security device or software that detects and actively prevents or removes malicious activity or items.
IPv4	Internet Protocol version v4. A communication protocol for identifying devices on a network. IPv4 addresses are 32-bit integers expressed in decimal notation.
IPv6	Internet Protocol version 6. A communication protocol that identifies and locates computers on networks, routing traffic across the Internet. IPv6 extends IP addresses from 32 bits to 128 bits.
Modems (modulator-demodulator)	A device that converts data into a format that is easy to transmit across a network. Once the data reaches its destination, the receiving modem converts it back to its original form.
Network Interface Cards (NICs)	A hardware that enables a computer or other device to connect to a network. NICs can be wired or wireless.
Network protocol	An established set of rules that determine how data is transmitted between different devices in the same network.
Network standards	Rules for data communications are needed for the interoperability of networking technologies and processes.
Node	A network-connected device that can send or receive information.
Open Systems Interconnection Model (OSI)	A conceptual framework used to describe the functions of a networking system. It is based on the concept of dividing a communication system into seven abstract layers.
Post Office Protocol version 3 (POP3)	An older protocol that was the standard for receiving email. Email is downloaded from an email server to a single device and then deleted from the server.
Remote Desktop Protocol (RDP)	A protocol used to remotely control another computer via a graphical user interface.
Repeater	A networking device that receives a signal amplifies it and retransmits it so it can cover longer distances or be received on the other side of an obstruction.
Router	A networking device that connects different networks or subnetworks and directs data packets between them.
Secure Shell (SSH)	A protocol is used to remotely control another device via a console or command shell securely with encryption.
Server	A powerful computer that stores files and applications. It can be accessed by other computers or devices on the same network as a central place to store information or use applications.

Term	Definition
Simple Message Transport Protocol (SMTP)	A protocol responsible for sending email.
Switch	A networking device that connects devices in a network. When a switch receives any data, it looks up the proper MAC address and forwards the data to the right device.
Teletype Network (Telnet)	A protocol used to remotely control another device via a console or command shell.
Transmission control protocol (TCP) protocol	An internet protocol that guarantees sent data makes it to its intended recipient. TCP guarantees that data reaches its intended recipient but is slower and resource-intensive.
Transmission Control Protocol/Internet Protocol Model (TCP/IP)	A set of standards that allow computers to communicate on a network. It is a four-layer model.
User Datagram Protocol (UDP)	An internet protocol (IP) used to send messages to other hosts on an IP network. UDP is fast and resource-efficient but doesn't guarantee that all packets will arrive.
Wireless access point (WAP)	A device that allows Wi-Fi devices to connect to a wired network. It serves as a central transmitter and receiver for wireless signals.
Wireless ad hoc network (WANET)	A type of local area network (LAN) that is created spontaneously to allow two or more wireless devices to connect with each other.
Wireless Internet Service Provider (WISP)	A cellular network that provides regional, nationwide, and global mesh coverage for mobile devices.
Wireless local-area network (WLAN)	A wireless network that connects computers and devices within homes, offices, or small businesses using wifi signals from routers, modems, and wireless access points. It uses IEEE 802.11 for home or office networks.
Wireless metropolitan area network (WMAN)	A wireless network that covers a larger geographical area. It uses IEEE 802.15 for city or campus networks.
Wireless Personal Area Network (WPAN)	A wireless network that connects devices within a short range (around 10 meters). WPANs use signals like Infrared, Zigbee, Bluetooth, and ultra-wideband. It uses use IEEE 802.15 for nearby devices.
Wireless wide area network (WWAN)	A wireless network that provides regional, nationwide, and global wireless coverage. This includes private networks of multinational corporations, the Internet, cellular networks like 4G, 5G, Long-Term Evolution (or LTE), and LoRaWAN.



Skills Network